

SHAY MORDECHAI

Offensive Security Researcher · Android Internals · Vulnerability Discovery

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SUMMARY

Security researcher focused on execution flow analysis and vulnerability discovery across runtime layers. Path: **AppSec** → **compiler internals** (CERT-IL #11265) → **Android** platform (Binder IPC, ART, SELinux). Currently mapping Android enforcement chains from Java API to Kernel.

TECHNICAL SKILLS

Research & Tools: Reverse Engineering · Android Internals · Linux Internals · Frida (DBI) · IDA Pro · JADX · Fuzzing · Native-Hook · Binder IPC · Memory Models (V8/JIT) · Threat Modeling · PAC/BTI (ARM64 Hardening)

Cryptographic protocols: TLS 1.2 implementation from RFC (PRF, key derivation, AES-CBC/HMAC-SHA256)

Languages: Python · C/C++ · Java · Assembly (x86 · ARM64)

PROJECTS & RESEARCH

Detailed write-ups at shay-mordechai.github.io

Android Binder IPC & Runtime Attack Surface Analysis (In Progress)

- Reversed Binder transaction flow from Java framework through libbinder to /dev/binder on ARM64; built Frida workflows to trace IPC calls and map execution boundaries during application bootstrap.
- Bypassed WMS Window Token requirement via raw Binder Parcel targeting SurfaceFlinger; blocked at SELinux AVC boundary (untrusted_app → surfaceflinger).

Android Application Execution & Malware Analysis (Digital Whisper · May 2026)

- Analyzed Anatsa banking trojan bootstrap: mapped execution flow through attachBaseContext() and ContentProvider.onCreate() before UI initialization; traced BaseDexClassLoader and LoadedApk paths with Frida.
- Documented ContentProvider as earliest native code execution vector in Android application lifecycle.
- Link: <https://shay-mordechai.github.io/blog/he/android-bootstrap-hijacking.html>

Compiler Security Research — SBCL/LISP Runtime (CERT-IL #11265 · Jul 2025)

- Identified three lexical scope isolation bypass vectors via semantic fuzzing of macro expansion; disclosed to CERT-IL (#11265) and MITRE.
- Built grammar-aware fuzzing harness generating AST-valid transformations to surface logical flaws without parser rejection.
- Link: <https://shay-mordechai.github.io/blog/he/compiler-sbcl-behavioral-research.html>

PROFESSIONAL EXPERIENCE

Independent Security Research | Jul 2025 – Present

- Three compiler vulnerabilities (SBCL lexical scope isolation) identified via AI-assisted semantic fuzzing; disclosed to CERT-IL (#11265) and MITRE.
- Android enforcement chain analysis: bypassed WMS token validation via raw Binder IPC; mapped SELinux AVC denial boundary (untrusted_app → surfaceflinger).
- Published Anatsa malware analysis in Digital Whisper; identified ContentProvider as pre-framework code execution vector.

Application Security Analyst | Israel Tax Authority | Aug 2024 – Jul 2025

- Led SAST/SCA operations (Checkmarx); mapped attack vectors and reduced false positives across production financial systems.
- Conducted threat modeling and execution flow analysis for a GCP-based file ingestion system; identified upload flow abuse scenarios and token handling gaps at design stage.
- Partnered with architects and R&D teams to translate OWASP findings into actionable remediation aligned with Secure SDLC.

EDUCATION

B.Sc. Computer Science | Jerusalem College of Technology | 2021 – 2025 · GPA: 89

Core Focus: Reverse Engineering · Malware Analysis · Network Protocols · Advanced OOP (Java, C++)